

Introduction to Python

What is Python?

Python is a high-level programming language known for its simple syntax and readability. It is commonly used for web development, automation, data analysis, artificial intelligence, and machine learning.

Variables

A variable is a name that refers to a value. Python variables do not require an explicit type declaration. For example, `age = 25` stores the integer 25 in the variable `age`.

Lists

A list is an ordered and mutable collection of items. Lists can contain values of different types. For example, `fruits = ['apple', 'banana', 'mango']` creates a list containing three strings.

Dictionaries

A dictionary stores data as key-value pairs. For example, `student = {'name': 'Rahul', 'age': 22}` stores a student's name and age. Dictionaries are mutable, so their contents can be changed after creation.

Functions

A function is a reusable block of code designed to perform a particular task. Functions are defined using the `def` keyword. They can accept parameters and can return a value.

Python and AI

Python is widely used in artificial intelligence because it has many libraries for machine learning, natural language processing, and data science. Popular libraries include NumPy, pandas, PyTorch, and scikit-learn.

Practice Questions

1. What is Python commonly used for?
2. What is a variable?
3. Are Python lists mutable?
4. How does a dictionary store data?
5. Why is Python popular for artificial intelligence?